

Q1 Matrix multiplication

10 marks

Matrix multiplication

Your Code

```
1 import java.util.*;
2 public class MatrixMultiplication{
3     public static void main(String[] args){
4         int [][]A={{1,2},
5                     {3,4}}
6     };
7     int [][]B={{5,6},{7,8}}
8     };
9     int rows=A.length;
10    int cols=B[0].length;
11    int common=B.length;
12    int [][]C=new int[rows][cols];
13    for(int i=0;i<rows;i++){
14        for(int j=0;j<cols;j++){
15            for(int k=0;k<common;k++){
16                C[i][j]+=A[i][k]*B[k][j];
17            }
18        }
19    }
20    System.out.println("Result matrix:");
21    for(int i=0;i<rows;i++){
22        for(int j=0;j<cols;j++){
23            System.out.print(C[i][j]+"");
24        }
25        System.out.println();
26    }
27 }
28 }
29 }
```

Input

stdin input

Output

```
Result matrix:
1024
4264
```

Q2 . Largest number

10 marks

. Largest number

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int[]arr={10,25,15,40,30};
6         int largest=arr[0];
7         for(int i=1;i<arr.length;i++){
8             if(arr[i]>largest){
9                 largest=arr[i];
10            }
11        }
12        System.out.println("largest number="+largest);
13    }
14 }
```

Input

stdin input

Output

```
largest number=40
```

 Run Save

Q3 Smallest number

10 marks

Smallest number

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[]args){
4         int []arr={12,34,56,32,89};
5         int smallest=arr[0];
6         for(int i=1;i<arr.length;i++){
7             if(arr[i]<smallest){
8                 smallest=arr[i];
9             }
10        }
11    }
12    System.out.println("smallest number="+smallest);
13 }
14 }
```

 Run Save**Input**

stdin input

Output

smallest number=12

Q4 . Nth largest number

10 marks

. Nth largest number

Your Code

```
1 import java.util.Arrays;
2 public class Main{
3     public static void main(String[]args){
4         int []arr={10,25,35,40,50};
5         int n=2;
6         Arrays.sort(arr);
7         System.out.println(n+"nth largest="+arr[arr.length-n]);
8     }
9 }
```

 Run Save**Input**

stdin input

Output

2nth largest=40

Q5 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```
1 public class Palindrome{
2     public static void main(String[] args){
3         String str="madam";
4         String rev="";
5         for(int i=str.length()-1;i>=0;i--){
6             rev=rev+str.charAt(i);
7         }
8         if(str.equals(rev))
9             System.out.println("palindrome");
10        else
11            System.out.println("not palindrome");
12    }
13 }
```

 Run Save**Input**

stdin input

Output

palindrome

Q6 Removing duplicates in array

10 marks

Removing duplicates in array

Your Code

```
1 import java.util.*;
2 public class HashSet{
3     public static void main(String[] args){
4         int []arr={1,2,2,3,4,5,6};
5         Set<Integer>set=new LinkedHashSet<>();
6         for(int num:arr){
7             set.add(num);
8         }
9         System.out.println("Array after removing duplicates:");
10        for(int num:set){
11            System.out.print(num+"");
12        }
13    }
14 }
```

 Run Save**Input**

stdin input

OutputArray after removing duplicates:
123456